F-DaC Forensics Data Carving Tool

Data Carving

Agenda

- Introduction - File Carving Overview
- Background - Terminology & Methodology
- File Carving challenges
- File Carving limits
- File Carving tools available
- F-DAC, The Forensic Data Carving tool
- In-Place (or Zero Storage) carving in CyberCheck4.0
- Conclusion

Introduction - File Carving Overview

Carving:

General term for extracting data (files) out of undifferentiated blocks (raw data)

File Carving:

Identifying and recovering files based on analysis of file formats

Introduction - File Carving Overview (contd..)

- File carving is a powerful technique because it can
  - Identify and recover files of interest from raw, deleted, damaged file system, memory, or swap space data
  - Assist in recovering files and data that may not be accounted for by the operating system and file system
  - Assist in simple data recovery
- Many file types have well-known values or magic numbers in the first few bytes of the file header
- Most file carvers
  - Identify specific types of file headers and/or footers
  - Carve out blocks between these two boundaries
  - Stop carving after a user-specified or set limit has been reached

Introduction - File Carving Overview (contd.)

File Carving Taxonomy:

Simson Garfinkel and Joachim Metz have proposed the following file carving taxonomy:

- Header/Footer Carving
- Header/Maximum (file) size Carving
- Block Based Carving
- Characteristic Based Carving
- Header/Embedded Length Carving
- File structure based Carving
- Semantic Carving
- Carving with Validation
- Fragment Recovery Carving

File Carving challenges

- Number of forensic cases growing exponentially today
- Investigations can be lengthy
  - – Machines tied up for days during investigations
  - – Forensic targets with GB or TB of storage
  - – Still need rapid turnaround, especially in time-sensitive cases involving potential loss of life or property -- think terrorists
- Not all file types have standard footer signature, determining end can be difficult
- Existing file carving tools typically produce many false positives and can miss key evidence
  - - Need file carving algorithms that identify more files and reduce the number of false positives
- Tools don't handle fragmentation very well

– A 'validator' can assist in testing the carved data

File Carving limits

- Unfortunately, not all file types have a standard footer signature, so determining the end can be difficult.
- Only one Carving method may not be enough,
  - May be time consuming

File Carving tools

- Foremost
- Scalpel
- DataLifter – Extractor Pro
- Encase
- FTK's Data Carving module
- CarvFs
- LibCarvPath
- PhotoRec
- PhotoRescue
- RevIt
- MagicRescue, etc.

File Carving tools (contd..)

- **Foremost**
  - ✓ Foremost is a console program to recover files based on their headers, footers, and internal data structures
  - ✓ Works on image files(dd, Safeback, Encase, etc), or directly on a drive.
  - ✓ Originally developed by the United States Air Force Office of Special Investigations and The Center for Information Systems Security Studies and Research .

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File Carving tools (contd..)

- Scalpel
  - ❖ Scalpel resulted from a complete rewrite of foremost 0.69, to enhance performance and decrease memory usage.

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F-DAC,
The Forensic Data Carving tool

- To optimize the search space, it searches the header signature at beginning of the sector.
- To search embedded files (ex: jpeg in doc, thumb nails in thumbs.db), it searches the header signature throughout the evidence .
- Provides a very good graphical user interface
- Generates a report file that gives detailed information (start offset, start sector, file size, etc) of carved files.

F-DAC,
The Forensic Data Carving tool

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F-DAC uses the following techniques for carving

- Header/Footer Carving
- Header/Maximum file size Carving
- Header/Embedded Length Carving
- File structure based Carving and
- Carving with Validation

F-DAC, The Forensic Data Carving tool

- It supports carving of following files
  - JPEG files
  - GIF files
  - BMP files
  - PNG files
  - PSD files
  - PDF files
  - ZIP files
  - HTML files
  - Compound files (DOC, PPT, EXCEL and Thumbs.db files)
  - Video files (AVI, DAT, MP4, MOV, WMV and 3gp)

F-DAC, The Forensic Data Carving tool

- It supports carving of following files
  - JPEG files
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In-Place (or Zero Storage) carving in CyberCheck4.0 (contd..)

- **Unallocated clusters**

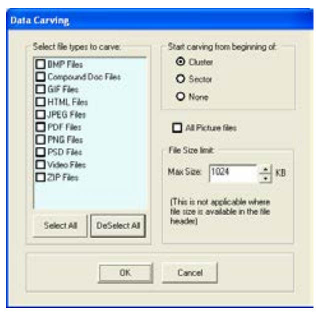
Cluster is defined as a logical unit of file storage on hard disk.

- These are the clusters, which are not allocated to any files on the hard disk according to File Management System.
- Unallocated clusters can result from deleting a file or formatting the partition (or entire disk).
- While deleting a file or formatting the hard disk the data that is there in the clusters remains unchanged. If the cluster resulted in this case contains data, it is called as used free cluster. Otherwise unused free or free cluster

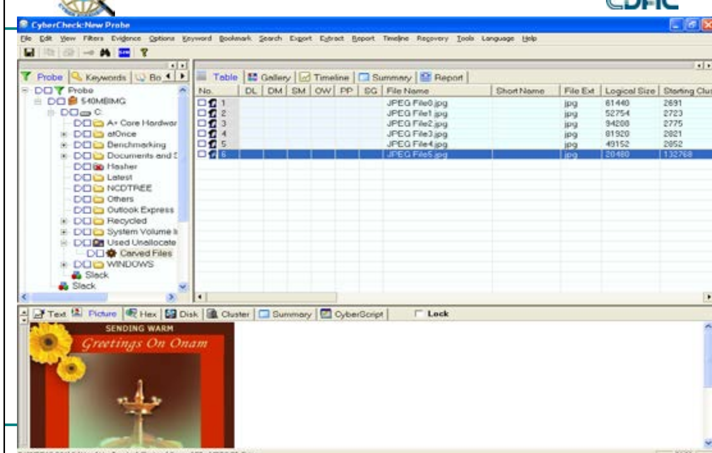
In-Place (or Zero Storage) carving in CyberCheck4.0 (contd..)

- **Lost clusters**
  - Lost clusters are the clusters, which are allocated to a file but are not having reference in the file allocation table.
  - In other words, a data fragment that does not belong to any file, according to the File Management System, and, therefore, is not associated with a file in the file allocation table.
  - Lost clusters can result from files not being closed properly, from shutting down a computer without first closing an application (power failure) or from ejecting a storage medium, such as a floppy disk, from the disk drive while the drive is reading or writing.

In-Place (or Zero Storage) carving in CyberCheck4.0 (contd..)



In-Place (or Zero Storage) carving in CyberCheck4.0 (contd..)



Conclusion

- Automation of fragmented file carving is difficult.
- Artificial Intelligence techniques can be used to locate next fragment.